Partial Visual-Tactile Fused Learning for Robotic Object Recognition

Tao Zhang, Yang Cong, *Senior Member, IEEE*, Jiahua Dong, and Dongdong Hou

***Abstract*—Currently, visual-tactile fusion learning for robotic object recognition has achieved appealing performance, due to the fact that visual and tactile data can offer complementary information. However: 1) the distinct gap between vision and touch makes it difficult to fully explore the complementary information, which would further lead to performance degradation and 2) most of the existing visual-tactile fused learning methods assume that visual and tactile data are complete, which is often difficult to be satisfied in many real-world applications. In this article, we propose a partial visual-tactile fused (PVTF) framework for robotic object recognition to address these challenges. Specifically, we first employ two modality-specific (MS) encoders to encode partial visual-tactile data into two incomplete subspaces (i.e., visual subspace and tactile subspace). Then, a modality gap mitigated (MGM) network is adopted to discover modality-invariant high-level label information, which is utilized to generate gap loss and further help updating the MS encoders for relatively consistent visual and tactile subspaces generation. In this way, the huge gap between vision and touch is mitigated, which would further contribute to mine the complementary visual-tactile information. Finally, to achieve data completeness and complementary visual-tactile information exploration simultaneously, a cycle subspace leaning technique is proposed to project the incomplete subspaces into a complete subspace by fully exploiting all the obtainable samples, where complete latent representations with maximum complementary information can be learned. A lot of comparative experiments conducted on three visual-tactile datasets validate the advantage of the proposed PVTF framework, by comparing with state-of-the-art baselines.**

***Index Terms*—Intelligent robots, partial multiview learning (PMVL), visual-tactile fused sensing (VTFS).**

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Tao Zhang, Jiahua Dong, and Dongdong Hou are with the State Key Laboratory of Robotics, Shenyang Institute of Automation and the Institute for Robotics and Intelligent Manufacturing, Chinese Academy of Sciences, Shenyang 110016, China, and also with the University of Chinese Academy of Sciences, Beijing 100049, China (e-mail: zhangtaosia@gmail.com; dongjiahua@sia.cn; houdongdong@sia.cn).

Yang Cong is with the State Key Laboratory of Robotics, Shenyang Institute of Automation and the Institute for Robotics and Intelligent Manufacturing, Chinese Academy of Sciences, Shenyang 110016, China (e-mail: congyang81@gmail.com).

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I. Introduction

As a fundamental problem in the research of intelligent robots, robotic object recognition has obtained tremendous improvement in recent years [1]. Intelligent robots need to recognize target objects before performing some certain actions [2]–[5]. The most popular method of recognizing a target object is to use its color images and then perform image classification [6]. However, practical manipulation tasks provide great challenges for object recognition tasks by only employing visual information. For instance, two glass bottles (their color and shape are same) filled with water in varying degrees cannot be distinguished easily only through visual features [7].

To that end, visual-tactile fused learning (VTFL) has been developed recently, which attracts lots of attention due to its encouraging performance. VTFL is inspired by human cognitive mechanisms [8], [9], which attempts to recognize visually similar objects by utilizing the complementary information provided by tactile modality [10], [11]. For example, Gao *et al.* [12] proposed a deep learning model to do attribute classification tasks, which is trained by fused visual and tactile data. They find that combining visual and tactile data leads to superior performance, in comparison to training the model only with visual or tactile data. Moreover, Liu *et al.* [7] found that kernel sparse coding is a good way to integrate preextracted visual and tactile features. They prove the effectiveness of their proposed method on a classification task of 18 household objects. All these work have proved that it is of great significance to carry out the related research on visual-tactile fused sensing (VTFS).

While VTFL methods are promising in robotic object recognition tasks [13], they suffer from the shortcoming. Existing VTFL methods usually make an assumption that both visual data and tactile data of each sample are accessible. Nevertheless, there often exists incomplete (or) partial scenarios where visual or tactile data are missing in many real-life situations. For instance, in robot grasping tasks, visual images taken by fixed cameras can be easily obscured by the manipulators or robot arms [7], [14], [15], which causes the visual modality unavailable. Moreover, noise, fault, and loss of signal in data collecting progress could also lead to the loss of instances, especially in the process of tactile data collection, since robots collect tactile data through continuous contact, which can be easily interrupted by outside disturbances.

To solve the missing data issues, a naive idea is to employ existing partial multiview learning (PMVL) methods [16]–[18]

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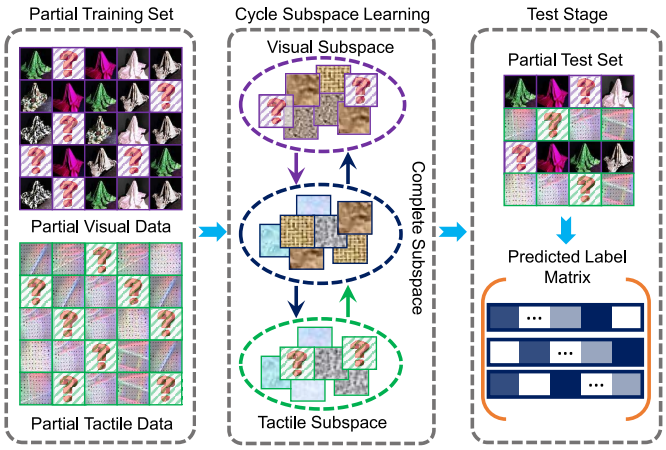

Fig. 1. Schematic of the proposed PVTF framework, which is designed to handle the robotic object recognition problem by fusing partial visual-tactile data. PVTF is capable of achieving data completion and visual-tactile complementary information exploration via cycle subspace learning.

or employ generative adversarial networks (GANs) [19] to synthesize the missing modality based on the other available modality [20], [21]. However, the gap across discriminant visual and tactile modalities is still the biggest obstacle to directly employ these methods. For instance, the reception field, signal types, and frequency of visual and tactile data are quite different. More specifically, the different data acquisition sensors bring great differences in frequency, where cameras take photos with relatively low frequency, whereas tactile sensors collect data by continuously touching with high frequency. Visual sensing emphasizes the perception of color and global shape, whereas tactile sensing focuses on perceiving hardness and contact force, leading to the big disparities of visual–tactile data in terms of data format and reception field. Naively applying existing PMVL or GANs methods would ignore the huge visual–tactile gap, and further make it difficult to excavate the complementary information, which inevitably leads to degradation of object recognition performance. Accordingly, a challenging problem arises: how to build a framework, which could effectively mitigate the visual–tactile gap and further explore complementary information to achieve accurate object recognition, when the input data are with arbitrary modalities or samples missing, which is an urgent problem to be solved in this article.

In order to address the above problems, we put forward a partial visual–tactile fused (PVTF) framework for robotic object recognition, as illustrated in Fig. 1. More details of PVTF are given in Fig. 2. To be more specific, two modality-specific (MS) encoders are used for encoding the input partial visual–tactile data into incomplete visual subspace and tactile subspace. By utilizing the modality-invariant high-level label information, a modality gap mitigated (MGM) network is introduced to establish the connection between the visual and tactile modalities, which is implicitly represented by a gap loss. The gap loss serves as a bridge to alleviate the gap between visual and tactile modalities, which supervises the generation of relatively consistent visual and tactile subspaces. Then, we adopt cycle encoding to complete missing data and explore visual–tactile complementary information simultaneously. On the one hand, the cycle encoding implicitly learns the combinations map of two incomplete visual and tactile subspaces to build a complete subspace, where the maximum complementary information is naturally obtained. On the other hand, it attempts to employ the latent representations in the complete subspace to reconstruct each available observation in the incomplete visual and tactile subspaces, which helps to preserve the original data structure distribution and further enhances robustness of the encoding process. Finally, we conduct abundant contrast experiment on three public datasets to illustrate the advantages of the proposed PVTF framework. Our contributions are listed as follows.

1) A PVTF is proposed for object recognition. As far as we know, PVTF is an earlier attempt for visual–tactile learning, which deals with the issue of partial data.
2) A simple yet effective MGM network is proposed to help consistent incomplete visual and tactile subspaces generation, where the visual–tactile gap is mitigated by discovering modality-invariant high-level label information.
3) A cycle subspace learning technique is encapsulated into our framework to seek a complete subspace, where missing instances are completed and the maximum visual–tactile complementary information can be naturally obtained.

The remainder of this article is organized as follows. Sections II and III introduces the related work and the PVTF framework, respectively. Sections IV and V give the details of our experiments and conclusion, respectively.

II. Related Work

We discuss the related work, i.e., VTFS for object recognition and PMVL in this section, following by introducing the differences across our approach and the existing methods.

A. Visual–Tactile Fused Sensing for Object Recognition

In the intelligent robots community, progressive progress researches have been performed on object recognition by fusion of visual and tactile data [13]. For instance, to solve household object recognition tasks, Liu *et al.* [7] presented a sparse coding-based model to explore mutual information among visual–tactile data. Sun *et al.* [22] proposed a systematic system for target identification and manipulation via visual data and tactile data. Liu *et al.* [23] attempted to retrieve images from tactile information, which is called as cross-modal matching. Then, they extend this model to zero-shot learning scenarios [24], which use the visual modality as a tool to help recognize untouched objects. Zheng *et al.* [25] proposed a lifelong learning to recognize surface materials by establishing the mechanism of knowledge forgetting from visual–tactile data.

Benefiting from the development of deep learning technologies [26], a variety of deep visual–tactile fused models is developed to achieve more accurate object recognition. For instance, Gao *et al.* [12] adopted deep neural networks to recognize household objects and perceive object properties by using both visual and tactile data. They demonstrate that the

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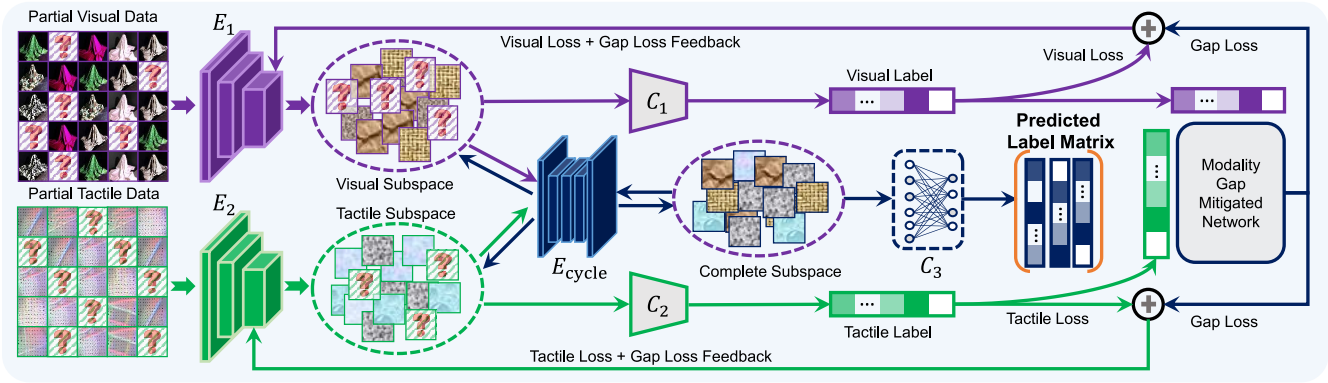

Fig. 2. Details of our PVTF framework. The partial (incomplete) visual–tactile data first go through encoders $E_1(\cdot)$ and $E_2(\cdot)$ for obtaining visual subspace $Z^{(1)}$ and tactile subspace $Z^{(2)}$, respectively. Two MS classifiers $C_1(\cdot)$ and $C_2(\cdot)$ are employed to predict visual label and tactile label. Then, the proposed MGM network is employed to generate gap loss, which helps updating $E_1(\cdot)$ and $E_2(\cdot)$ by utilizing modality-invariant high-level label information, and further contributes to visual–tactile gap mitigation. The cycle encoder $E_{\text{cycle}}(\cdot)$ fully explores the complementary visual–tactile information and then generate a complete subspace, where complete latent representations for the input objects can be obtained. Finally, the learned complete latent representations are fed to the final classifier $C_3(\cdot)$ to get the final predictions.

visual information and tactile information are indeed complementary. Zheng *et al.* [27] developed a fusion network for surface material perception, which utilizes the max-voting and random sampling strategy to explore the complementary information contained in the visual–tactile data. Yuan *et al.* [28] employed a multiinput network to connect visual and tactile properties. They also demonstrate that the tactile modality really "helps" visual recognition. Lee and Lee *et al.* [20], Li *et al.* [21] adopted a generative adversarial learning-based prediction model for producing fake visual images (or tactile outputs) from tactile input (or visual images), which can be further used to train classification models. Zhang *et al.* [29], [30] made an early attempt for unsupervised visual and tactile fusion clustering.

However, all the above methods make an unrealistic assumption that both visual and tactile data are complete, which would undoubtedly cause performance degradation or recognition failure. Thus, in this article, we focus on building a visual–tactile fused model that can solve the partial data cases.

B. Partial Multiview Learning

As an important branch of machine learning, PMVL methods usually transform partial input data cases into complete multiview learning tasks. For instance, Yuan *et al.* [16] first trained multiple view-specific classifiers for each view data, where each view data can be expressed by prediction scores. Then, they use existing prediction scores to establish a multiview integration framework to estimate the missing prediction scores, and further achieve multiview prediction. Tran *et al.* [17] imputed the missing modalities by a cascaded residual autoencoder, which leverages the relatedness among different modalities. They provide an effective data imputation way that leverages strengths of residual learning and autoencoder networks. Zhang *et al.* [18] encoded the partial multiview data into a complete subspace through mimicking data transmitting, which fully excavates any obtainable samples and views to generate complete latent representations. Wang *et al.* [31] encapsulated adversarial learning into

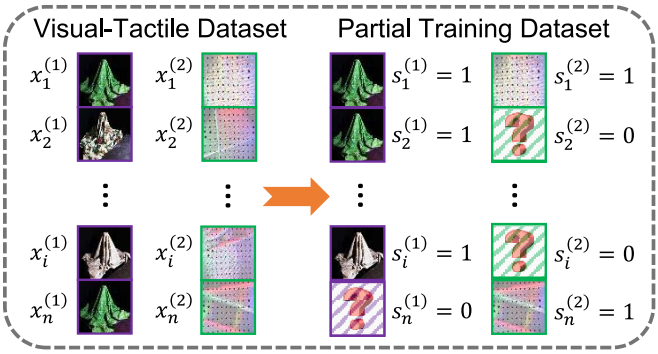

Fig. 3. Illustration of the process of obtaining the partial visual–tactile training dataset.

their model to complete the missing modality conditionally based on the other obtainable modality. Although their method achieves promising results in modeling the partial data, it needs at least one complete modality, which hinders its universality. Wen *et al.* [32] proposed a unified embedding alignment framework (UEAF) for incomplete multiview learning, which introduces a locality-preserved reconstruct term and a consensus graph to infer the missing views.

However, these methods mentioned above do not explicitly consider the huge gap between visual and tactile modalities, which would lead to the complementary information not be well exploited.

III. Proposed Framework

A. Preliminaries

We denote a complete paired visual–tactile training dataset with n samples as $X = \{x_1^{(v)}, \ldots, x_n^{(v)}\}_{v=1}^{2}$, and the label of the nth sample is denoted as y_n. The nth sample of the vth modality is represented as $x_n^{(v)}$, where $v = 1$ and $v = 2$ represent the visual modality and tactile modality, respectively. As shown in Fig. 3, we randomly select some samples from the visual–tactile dataset to simulate the missing sample points, i.e., the incomplete data cases. We denote $s_i^{(1)}$ as the indicator of the

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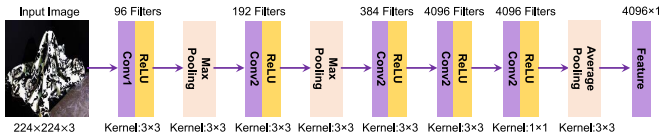

Fig. 4. Details of the proposed 5-layer visual encoder $E_1(\cdot)$ in Fig. 2, which includes five convolution operations, two max pooling operations, and a average pooling operation.

availability for the ith ($i < n$) sample in the visual modality, where $s_i^{(1)} = 0$ denotes the ith sample in visual modality is missing and $s_i^{(1)} = 1$ denotes the ith sample is not missing. Similarly, $s_i^{(2)} = 0$ denotes the ith sample in tactile modality is missing and $s_i^{(2)} = 1$ denotes the ith sample is not missing. In this way, we obtain the partial visual–tactile training dataset X_s ($X_s \subseteq X$). PVTF aims to train the framework by using the partial visual–tactile training dataset X_s, and classifies a new sample with arbitrary possible modality-missing patterns.

B. Visual–Tactile Gap Mitigation

To better explore the complementary information between visual and tactile modalities, we need to mitigate the huge visual–tactile gap first. Different from existing visual–tactile fused object recognition methods [12], [27], which either learn the score weights of each modality or try to fuse the visual–tactile features in low-level feature space, we take advantage of the modality-invariant high-level label information to bridge the visual–tactile gap. Since the visual data and tactile data originating from an object have the same label, the label information naturally establishes the connection between vision and touch.

To this end, we first use the visual encoder $E_1(\cdot)$ to encode all the samples in the partial training dataset X_s into the incomplete visual subspace $Z^{(1)}$, where the subspace representations $z_i^{(1)}$ of the ith instance in the visual modality are able to be learned. The details of $E_1(\cdot)$ are shown in Fig. 4, which projects each input image to an eigenvector $z_i^{(1)}$, which could further be used to generate visual subspace $Z^{(1)}$. Similarly, the tactile input of the ith sample is fed to the tactile encoder $E_2(\cdot)$ to learn the tactile feature representations $z_i^{(2)}$, and further generates the incomplete tactile subspace $Z^{(2)}$. The structures of $E_2(\cdot)$ are similar to $E_1(\cdot)$. The objective function of $E_1(\cdot)$ and $E_2(\cdot)$ is given as

$$\mathcal{L}_{E_1} = \mathcal{L}_1 + \lambda_1 \mathcal{L}_{\text{Gap}}, \quad \mathcal{L}_{E_2} = \mathcal{L}_2 + \lambda_2 \mathcal{L}_{\text{Gap}} \tag{1}$$

where $\mathcal{L}_1$, $\mathcal{L}_2$, and $\mathcal{L}_{\text{Gap}}$ are the visual loss, tactile loss, and gap loss, respectively. More specifically, $\mathcal{L}_1$ and $\mathcal{L}_2$ are given as

$$\begin{aligned} \mathcal{L}_1 &= \frac{1}{\sum_{i=1}^{n} s_i^{(1)}} \sum_{i=1}^{n} s_i^{(1)} \left\| y_i - y_{\text{pre}_i}^{(1)} \right\|_2^2 \\ \mathcal{L}_2 &= \frac{1}{\sum_{i=1}^{n} s_i^{(2)}} \sum_{i=1}^{n} s_i^{(2)} \left\| y_i - y_{\text{pre}_i}^{(2)} \right\|_2^2 \end{aligned} \tag{2}$$

where $y_{\text{pre}_i}^{(1)} = C_1(z_i^{(1)})$ is the predicted visual label, and $y_{\text{pre}_i}^{(2)} = C_2(z_i^{(2)})$ is the predicted tactile label. λ_1 and λ_2 are the tradeoff parameters, which balance the effect of the gap loss. $C_1(\cdot)$ is the visual classifier and $C_2(\cdot)$ is the tactile classifier, which are implemented by two softmax classifiers.

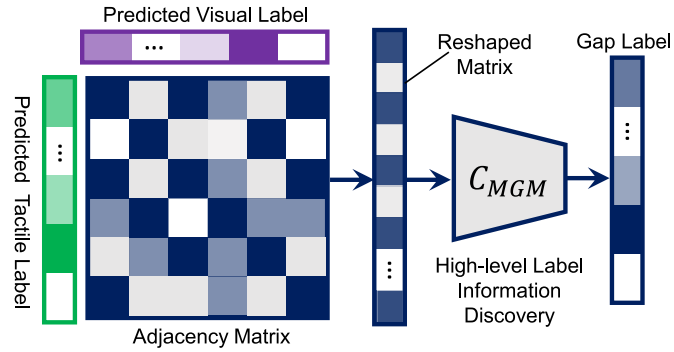

Fig. 5. Details of the proposed MGM network.

The losses $\mathcal{L}_1$ and $\mathcal{L}_2$ could regularize the encoders $E_1(\cdot)$ and $E_2(\cdot)$ to learn more distinguishable feature subspace $Z^{(1)}$ and $Z^{(2)}$ for different objects, respectively.

$\mathcal{L}_{\text{Gap}}$ is the shared loss, which helps updating the visual encoder $E_1(\cdot)$ and tactile encoder $E_2(\cdot)$ at the same time. Instead of naively averaging or weighting the MS classification scores, we explore the initial scores and discover the correlations across different modalities. The gap loss $\mathcal{L}_{\text{Gap}}$ is generated by the proposed MGM network. The details of the MGM network are given in Fig. 5. After getting the initial predicted visual labels $\{y_{\text{pre}_i}^{(1)}\}_{i=1}^{n}$ and the tactile labels $\{y_{\text{pre}_i}^{(2)}\}_{i=1}^{n}$ for all the instances in the visual subspace $Z^{(1)}$ and tactile subspace $Z^{(2)}$, we choose the instances which appear in the visual subspace and the tactile subspace simultaneously to build a set $\{y_{\text{pre}_k}^{(1)}, y_{\text{pre}_k}^{(2)}, y_k^{(3)}\}_{k=1}^{K}$. To be more specific, $\{y_{\text{pre}_k}^{(1)}\}_{k=1}^{K} \subseteq \{y_{\text{pre}_i}^{(1)}\}_{i=1}^{n}$ and $\{y_{\text{pre}_k}^{(2)}\}_{k=1}^{K} \subseteq \{y_{\text{pre}_i}^{(2)}\}_{i=1}^{n}$ are the initial predictions corresponding the kth instances from the visual subspace $Z^{(1)}$ and the tactile subspace $Z^{(2)}$. K is number of these instances and $y_k^{(3)}$ is the label of the kth instance.

Then, we do a transformation from the two MS subspace predictions (i.e., $y_{\text{pre}_k}^{(1)}$ and $y_{\text{pre}_k}^{(2)}$), to construct a cross-modality label-level adjacency matrix $\mathbf{a}_k$ by multiplying $y_{\text{pre}_k}^{(1)}$ and the transpose of $y_{\text{pre}_k}^{(2)}$, i.e., $\mathbf{a}_k = y_{\text{pre}_k}^{(2)} \cdot (y_{\text{pre}_k}^{(1)})^\top$. By this way, the elements in $\mathbf{a}_k$ are the multiplication of the pairwise predicted scores. Then, we reshape the matrix $\mathbf{a}_k$ to a vector, and feed it to the proposed MGM network $C_{\text{MGM}}(\cdot)$ to reveal the hidden correction between the two subspaces, and further generate the gap label $C_{\text{MGM}}(y_{\text{pre}_k}^{(2)} \cdot (y_{\text{pre}_i}^{(1)})^\top)$. The gap loss $\mathcal{L}_{\text{Gap}}$ can be written as

$$\mathcal{L}_{\text{Gap}} = \frac{1}{K} \sum_{k=1}^{K} \left\| y_k^{(3)} - C_{\text{MGM}}\left(y_{\text{pre}_k}^{(2)} \cdot \left(y_{\text{pre}_i}^{(1)} \right)^\top \right) \right\|_2^2. \tag{3}$$

$C_{\text{MGM}}(\cdot)$ is a two-layer fully connected network with LeakyReLU activation [33] deployed in the first layer. In this way, modality-invariant high-level label information is introduced to supervise the generation of relatively consistent subspace $Z^{(1)}$ and $Z^{(2)}$, where the large differences among visual data and tactile data can be alleviated.

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C. Missing Data Completion and Visual–Tactile Complementary Information Exploration

Considering the challenge of completing the missing data and exploring the complementary information among visual and tactile modalities, we aim to learn a flexible mapping to project the incomplete subspace $Z^{(1)}$ and $Z^{(2)}$ into a complete subspace $Z^{(3)}$. The maximum complementary visual–tactile information can further be obtained in $Z^{(3)}$. We first give the definition of the complete subspace $Z^{(3)}$ as follows.

Definition 1: If each MS subspace $Z^{(1)}$ or $Z^{(2)}$ can be reconstructed from $Z^{(3)}$ by a mapping $\{Z^{(1)}, Z^{(2)}\} = E_{\text{cycle}}(Z^{(3)}, \Theta_1)$, where Θ_1 are parameters governing the mapping function, then $Z^{(3)}$ is a complete subspace.

Intuitively, we can reconstruct each MS subspace (i.e., $Z^{(1)}$ and $Z^{(2)}$) in a numerically stable way. According to [34], given observations $Z^{(1)}$ and $Z^{(2)}$, each subspace is conditionally independent to the given complete subspace. Thus, we can model the likelihood with respect to $Z^{(3)}$ as

$$\begin{aligned} p\left(Z^{(1)}|Z^{(3)}\right) &\propto \exp^{-\Delta\left(Z^{(1)}, E_{\text{cycle}}\left(Z^{(3)}, \Theta_1\right)\right)} \\ p\left(Z^{(2)}|Z^{(3)}\right) &\propto \exp^{-\Delta\left(Z^{(2)}, E_{\text{cycle}}\left(Z^{(3)}, \Theta_1\right)\right)} \end{aligned} \tag{4}$$

where $\Delta(Z^{(1)}, E_{\text{cycle}}(Z^{(3)}, \Theta_1))$ is the loss for generating visual subspace, and $\Delta(Z^{(2)}, E_{\text{cycle}}(Z^{(3)}, \Theta_1))$ is the loss for generating tactile subspace.

The mapping $E_{\text{cycle}}(Z^{(3)}, \Theta_1))$ attempts to seek a complete subspace $Z^{(3)}$, where $Z^{(3)}$ consists of $\{z_i^{(3)}\}_{i=1}^n$ and $z_i^{(3)}$ denotes the ith latent representation. Similarly, the visual subspace consists of $\{z_i^{(1)}\}_{i=1}^n$, and the tactile subspace consists of $\{z_i^{(2)}\}_{i=1}^n$

According to the independent and identically distributed (IID) theorem, we could deduce the logarithmic-likelihood equation as follows:

$$\begin{aligned} &\mathcal{L}_{E_{\text{cycle}}}\left(\left\{z_i^{(3)}\right\}_{i=1}^n, \Theta_1\right) \\ &\quad = \sum_{i=1}^n \left(\ln p\left(z_i^{(1)}|z_i^{(3)}\right) + \ln p\left(z_i^{(2)}|z_i^{(3)}\right)\right) \\ &\propto -\sum_{i=1}^n \left(\Delta\left(z_i^{(1)}, E_{\text{cycle}}\left(z_i^{(3)}, \Theta_1\right)\right)\right. \\ &\qquad \left. + \Delta\left(z_i^{(2)}, E_{\text{cycle}}\left(z_i^{(3)}, \Theta_1\right)\right)\right). \end{aligned} \tag{5}$$

Since we seek complete latent representations that could recover the partial (available) observations, we introduce a cycle schema to specialize $\Delta(\cdot)$ and obtain the following loss:

$$\begin{aligned} \mathcal{L}_{E_{\text{cycle}}}\left(\left\{z_i^{(3)}\right\}_{i=1}^n, \Theta_1\right) = \sum_{i=1}^n &\left(s_i^{(1)}\right)\left\|E_{\text{cycle}}\left(z_i^{(3)}, \Theta_1\right)\right) - z_i^{(1)}\right\|_2^2 \\ &+ \left(s_i^{(2)}\right)\left\|E_{\text{cycle}}\left(z_i^{(3)}, \Theta_1\right)\right) - z_i^{(2)}\right\|_2^2. \end{aligned} \tag{6}$$

In this way, $E_{\text{cycle}}(\cdot)$ encodes comprehensive information from diverse accessible modalities and instances into complete representations $z_i^{(3)}$, which implicitly ignore the data missing patterns. $E_{\text{cycle}}(\cdot)$ is implemented with a two-layer fully connected network in this article. Since $z_i^{(3)}$ is the complete subspace, it should be complete compared with incomplete representations $z_i^{(1)}$ and $z_i^{(2)}$. We give the following theorem, which guarantees the completeness of $Z^{(3)}$, followed by a theoretical proof.

Theorem 1: $\forall z_i^{(1)}, z_i^{(2)}$, if $z_i^{(3)}$ are complete latent representations, there must exist a mapping $\Psi : z_i^{(1)} = \Psi(z_i^{(3)})$ and $z_i^{(2)} = \Psi(z_i^{(3)})$.

Proof: According to (6), there exists $z_i^{(1)} = E_{\text{cycle}}(z_i^{(3)}, \Theta_1))$ and $z_i^{(2)} = E_{\text{cycle}}(z_i^{(3)}, \Theta_1))$, where $E_{\text{cycle}}(\cdot)$ is the mapping from $z_i^{(3)}$ to $z_i^{(1)}$ and $z_i^{(2)}$. Thus $\forall z_i^{(1)}, z_i^{(2)}$, there exists a mapping $E_{\text{cycle}}(\cdot)$, which proves the completeness of $Z^{(3)}$. ■

Noticing: In order to project multimodal eigenvectors into a complete latent subspace, most existing methods usually employ the following schema: $\mathcal{L}_{E_{\text{cycle}}} = \sum_{i=1}^n \|E_{\text{cycle}}(z_i^{(1)}, \Theta_1)) - z_i^{(3)}\|_2^2 + \|E_{\text{cycle}}(z_i^{(2)}, \Theta_1)) - z_i^{(3)}\|_2^2$. This equation reveals that we can employ multiple single modality to reconstruct a complete subspace. However, these methods are not reasonable to maximize the multiple complementary information, especially for the missing data cases. Different from traditional methods that bluntly fuse the multimodality information, we employ an alternative route, i.e., $\mathcal{L}_{E_{\text{cycle}}} = \sum_{i=1}^n \|E_{\text{cycle}}(z_i^{(3)}, \Theta_1)) - z_i^{(1)}\|_2^2 + \|E_{\text{cycle}}(z_i^{(3)}, \Theta_1)) - z_i^{(2)}\|_2^2$, which learns complete representations that implicitly explore the multiple complementary information. Our model makes an assumption that the information from different modalities are originated from the latent complete representations $z_i^{(3)}$, and they can be mapped to each individual modality subspace. On the basis of information theory, the cycle encoder is able to be regarded as a communication channel, i.e., $p(Z^{(1)}|Z^{(3)})$ and $p(Z^{(2)}|Z^{(3)})$, which could be mimicked as data transmitting. Compared with traditional methods, our model is better at exploring complete information, especially in data missing scenarios.

After getting the complete latent representations $z_i^{(3)}$, we train the classifier $C_3(\cdot)$ to predict labels for these samples. $C_3(\cdot)$ is implemented with the following classification loss:

$$\mathcal{L}_{C_3} = \Delta\left(y_i, y_{\text{pre}_i}\right) = \left\|y_i - C_3\left(z_i^{(3)}, \Theta_2\right)\right\|_2^2 \tag{7}$$

where $C_3(z_i^{(3)}, \Theta_2) = \arg\max_{y_{\text{pre}_i}} \mathbb{E}_{z_i^{(3)} \sim \mathcal{T}(y_{\text{pre}_i})} F(\mathbf{z}^{(3)}, z_i^{(3)})$. $F(\mathbf{z}^{(3)}, z_i^{(3)}) = \phi(\{z_i^{(3)}\}_{i=1}^n, \Theta_2)^\top \phi(z_i^{(3)}, \Theta_2)$, where $\phi(\cdot, \Theta_2)$ is a feature mapping function for $\{z_i^{(3)}\}_{i=1}^n$. $\mathcal{T}(y_{\text{pre}_i})$ are a set of latent representations from class y_i, and y_{pre_i} is the label predicted by $C_3(\cdot)$. Θ_2 are the parameters for the classifier $C_3(\cdot)$. We set $\phi(z_i^{(3)}, \Theta_2) = z_i^{(3)}$ in this article for simplicity and effectiveness. To prevent from the loss $\mathcal{L}_{C_3}$ being zero, we further modify $\mathcal{L}_{C_3}$ as follows:

$$\begin{aligned} \mathcal{L}_{C_3} = \max_{y_{\text{pre}_i}} \Big(0, \Delta\left(y_i, y_{\text{pre}_i}\right) &+ \mathbb{E}_{z_i^{(3)} \sim \mathcal{T}(y_{\text{pre}_i})} F\left(\mathbf{z}^{(3)}, z_i^{(3)}\right) \\ &- \mathbb{E}_{z_i^{(3)} \sim \mathcal{T}(y_i)} F\left(\mathbf{z}^{(3)}, z_i^{(3)}\right)\Big). \end{aligned} \tag{8}$$

Different from the widely used cross-entropy loss, the introduced loss $\mathcal{L}_{C_3}$ penalizes the misclassification samples. For example, if an instance is rightly classified, i.e., $y_i = y_{\text{pre}_i}$, then $\mathcal{L}_{C_3}$ is zero. On the contrary, if an instance is wrongly

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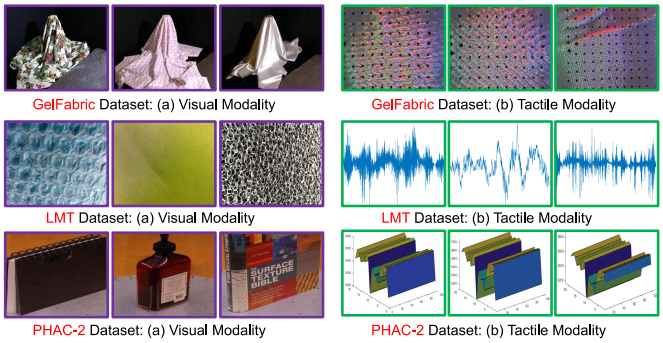

Fig. 6. Representative color images and tactile signals of the employed GelFabric dataset [28], LMT dataset [38], [39], and PHAC-2 dataset [12]. From these visualized pictures, we can intuitively see that there exists big gap between visual and tactile modalities. It should be noted that we merely visualize the electrode impedance component of the PHAC-2 dataset for convenience, even though there exists multiple types of signals. For more details, we can check [12], [28], [38], and [39]. It is worth noting that the tactile signals in the PHAC-2 dataset consist of multiple components. We only visualize the electrode impedance component for simplicity. More details of these datasets can be found in their corresponding references.

classified, i.e., $y_i \neq y_{\text{pre}_i}$, this equation would increase the similarity between $z_i^{(3)}$ and the center of a instance whose label is y_i. Obviously, (8) naturally leads to more separable representations. Thus, we employ the loss $\mathcal{L}_{C_3}$ to train the final classifier $C_3(\cdot)$. From what has been discussed above, the total loss function of our proposed PVTF framework can be summarized as follows:

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{E_1} + \mathcal{L}_{E_2} + \mathcal{L}_{E_{\text{cycle}}}\left(\left\{z_i^{(3)}\right\}_{i=1}^{n}, \Theta_1\right) + \mathcal{L}_{C_3}. \quad (9)$$

D. Details of Training our PVTF Framework

Our PVTF framework is trained by the following steps, which are summed up as follows.

Setp 1 (Initialization): We randomly select a incomplete visual–tactile training dataset X_s from the given complete paired visual–tactile training dataset X, as shown in Fig. 3. All the network parameters are initialized by the Xavier initializer [35].

Setp 2 (Training MS Encoders and Classifiers): Losses $\mathcal{L}_{E_1}$ and $\mathcal{L}_{E_2}$ are calculated by (1). Then $\mathcal{L}_{E_1}$ and $\mathcal{L}_{E_2}$ are employed to update their corresponding encoders. Meanwhile, the visual classifier $C_1(\cdot)$ and tactile classifier $C_2(\cdot)$ are updated by the losses $\mathcal{L}_1$ and $\mathcal{L}_2$. We use Adam [36] optimizers in this article.

Setp 3 (Training the Cycle Encoder and the Final Classifier): In this step, we employ the cycle encoder loss $\mathcal{L}_{E_{\text{cycle}}}(\{z_i^{(3)}\}_{i=1}^{n}, \Theta_1)$ to update the proposed cycle encoder by sending it to an Adam optimizer. Then, the final classifier $C_3(\cdot)$ is optimized by the loss $\mathcal{L}_{C_3}$.

Setp 4: Once the proposed PVTF framework is well trained, a test sample with arbitrary modality-missing patterns is fed into our PVTF framework to predict its label. The batch size is set to 64, and our framework is implemented by Tensorflow 1.12.0 [37].

In Algorithm 1, the overall algorithm of the proposed PVTF framework is summarized.

Algorithm 1 Algorithm for the PVTF Framework

Input: Complete paired visual-tactile dataset: X and its label: y_n. The maximum iterations: MaxIter. The trade-off parameters: λ_1 and λ_2.
Initialization: Obtaining the partial visual-tactile training dataset: X_s. Initializing all the network parameters with the Xavier initializer [35].
Training:
for iter $\leq$ MaxIter **do**
  Training modality-specific encoders $E_1(\cdot)$ and $E_2(\cdot)$ by their corresponding losses $\mathcal{L}_{E_1}$ and $\mathcal{L}_{E_2}$;
  Training modality-specific classifiers $C_m(\cdot)$ with corresponding losses $\mathcal{L}_m$, $\forall m = 1, 2$;
  Training $E_{cycle}(\cdot)$ with $\mathcal{L}_{E_{cycle}}(\{z_i^{(3)}\}_{i=1}^{n}, \Theta_1)$;
  Training the final classifier $C_3(\cdot)$ with $\mathcal{L}_{C_3}$;
end for
Training:
Giving a test sample with arbitrary modality-missing patterns.
Classifying the test sample with the PVTF framework;
return Predicted label.

IV. Experiments

The settings of our experiments, including a brief introduction of the used datasets, comparison methods, and evaluation metrics, are provided in this section. We also evaluate the effectiveness of the PVTF framework on three public available visual–tactile datasets.

A. Settings of Our Experiments

1) Brief Introduction of the Used Datasets: Three well-known visual–tactile datasets (namely, *GelFabric* [28], *LMT* [38], [39], and *PHAC-2* [12]) are used for verifying the advantages of our PVTF framework,as shown in Fig. 6. Details of the used datasets are introduced as follows.

GelFabric dataset consists of 1190 color images taken by a Canon T2i SLR camera under different imaging conditions, where ten color images are taken for each fabric. Meanwhile, a widely used tactile sensor (i.e., Gelsight [40]) is employed to collect 1190 tactile images, where ten tactile images are captured for each fabric. We resize both tactile and color images to 224×224. LMT dataset has 108 different surface material instances, where each material instance has ten color images and 30 tactile acceleration traces. The first ten tactile signals and color images are employed to construct the used visual–tactile dataset in the experiments. We use all the color images and the first ten tactile acceleration traces to construct the paired visual–tactile dataset in this article. All the color images are resized to 224×224, and the tactile signals are converted to spectrogram images similar with [25], which are further fed to the tactile encoder. PHAC-2 dataset has 53 objects, which contains the most frequent objects in the household. They capture eight color images for each object, and all the color images are resized to 224×224. In addition, they capture ten tactile signals for each object. We use the first eight tactile signals,

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TABLE I
OBJECT RECOGNITION PERFORMANCE (%) ON THE GELFABRIC DATASET, WHEN THE MISSING RATE MR IS SET TO 0.1. THE BEST PERFORMANCE IS HIGHLIGHTED BY RED, AND THE SECOND BEST PERFORMANCE IS HIGHLIGHTED BY GREEN

Methods		Accuracy	Precision	F1-Score
MS	Vision [42]	88.28 ± 1.21	88.20 ± 1.57	94.20 ± 1.66
	Touch [42]	70.31 ± 1.26	77.44 ± 1.32	68.73 ± 1.38
VTFS	DLTU [12]	89.06 ± 0.84	89.01 ± 1.37	88.31 ± 1.12
	DLSMC [27]	88.28 ± 0.89	87.45 ± 1.17	87.27 ± 1.83
PMVL	UEAF [32]	43.19 ± 0.06	67.96 ± 0.26	45.96 ± 0.04
	CPMNET [18]	78.15 ± 0.90	78.54 ± 0.12	75.45 ± 1.04
Ours		97.48 ± 0.86	98.32 ± 0.96	97.31 ± 1.12

TABLE II
OBJECT RECOGNITION PERFORMANCE (%) ON THE LMT DATASET, WHEN THE MISSING RATE MR IS SET TO 0.1. THE BEST PERFORMANCE IS HIGHLIGHTED BY RED, AND THE SECOND BEST PERFORMANCE IS HIGHLIGHTED BY GREEN

Methods		Accuracy	Precision	F1-Score
MS	Vision [42]	81.25 ± 1.25	87.83 ± 1.87	82.64 ± 1.03
	Touch [42]	82.03 ± 1.31	85.03 ± 1.23	82.07 ± 1.37
VTFS	DLTU [12]	63.28 ± 1.76	64.68 ± 1.06	62.37 ± 1.29
	DLSMC [27]	81.64 ± 1.27	83.72 ± 0.99	81.97 ± 1.45
PMVL	UEAF [32]	47.78 ± 0.00	74.09 ± 0.06	50.56 ± 0.04
	CPMNET [18]	70.37 ± 1.88	75.55 ± 1.65	69.46 ± 1.95
Ours		90.47 ± 0.22	94.73 ± 0.17	91.53 ± 0.75

TABLE III
OBJECT RECOGNITION PERFORMANCE (%) ON THE PHAC-2 DATASET, WHEN THE MISSING RATE MR IS SET TO 0.1. THE BEST PERFORMANCE IS HIGHLIGHTED BY RED, AND THE SECOND BEST PERFORMANCE IS HIGHLIGHTED BY GREEN

Methods		Accuracy	Precision	F1-Score
MS	Vision [42]	78.91 ± 1.24	88.26 ± 1.86	77.84 ± 1.05
	Touch [42]	67.19 ± 1.31	81.51 ± 1.23	69.31 ± 1.37
VTFS	DLTU [12]	60.94 ± 0.75	68.10 ± 1.06	61.15 ± 0.95
	DLSMC [27]	76.56 ± 1.06	83.99 ± 1.16	75.73 ± 0.93
PMVL	UEAF [32]	40.56 ± 0.01	63.20 ± 0.06	43.63 ± 0.03
	CPMNET [18]	81.13 ± 1.65	78.43 ± 1.40	78.12 ± 1.12
Ours		88.06 ± 0.20	93.55 ± 0.15	89.25 ± 0.10

and then the tactile signals are preprocessed in a similar way with [12].

Partial Visual–Tactile Dataset Construction: Similar to PMVL methods [18], [41], given the number of samples in visual–tactile dataset as n, the definition of missing rate could be given as $\mathrm{MR} = \tilde{n}/n$, where $\tilde{n}$ is the number of the randomly picked samples from the tactile dataset and the visual dataset, respectively.

2) Comparison Methods and Evaluation Metrics: We compare our proposed PVTF framework with six representative baselines to illuminate its superiority. Specifically, two MS-based methods, two VTFS-based methods, and two PMVL-based methods are employed in this article.

MS-Based Methods: Vision performs object recognition by using the visual data to train a AlexNet [42] network. *Touch* uses the modified AlexNet network to classify tactile data. Specifically, we adjust the AlexNet [42] to conform to the special tactile data format of the PHAC-2 and LMT dataset.

VTFS-Based Methods: DLTU [12] is an earlier work that employs deep neural networks for VTFS. DLTU achieves encouraging recognition results compared with hand-designed features-based methods. *DLSMC* [27] employs deep learning technique to learn visual features and tactile features, respectively, and then fuse these features by randomly sampling and concatenation.

PMVL-Based Methods: UEAF [32] introduces a locality-preserved reconstruction term to infer the missing views such that all views can be naturally aligned. It also employs a consensus graph to guarantee the common local structure of multiple views and in turn can further align the incomplete views and inferred views. *CPMNET* [18] proposes a novel PMVL-based algorithm to handle arbitrary view-missing patterns by jointly exploit all available views and samples.

Referring to [43], three widely used classification metrics, i.e., *accuracy*, *precision*, and *F1-Score*, are used to prove the advantages of our PVTF framework. For all metrics, higher value means better model performance.

B. Experimental Results

For a fair comparison, the parameters of the compared methods mentioned above are tuned by following the setting in the corresponding papers. Notice that the missing data are randomly selected, and the experimental results would fluctuate to some extent.

Tables I–III present the object recognition performance on the GelFarbric, LMT, and PHAC-2 datasets, respectively, where we set MR to be 0.1. Generally, we can obtain the following observations.

1) Our PVTF framework always achieve the best recognition performance, which reveals the effectiveness of our approach. Taking the results on the GelFabric dataset as an example, we raise the performance bar by nearly 8.42%, 9.31%, and 3.11% in terms of accuracy, precision, and F1-Score, respectively, compared with the second best methods.
2) Comparing "Ours" with MS-based methods (e.g.,, "Vision," "Touch"), Ours achieves better performance. For instance, compared with Vision, Ours raise the performance bar by nearly 9.20%, 10.12%, and 3.11% in terms of accuracy, precision, and F1-Score, respectively, on the GelFabric dataset. The results demonstrate that the visual information and tactile information are indeed complementary, and properly fusing visual–tactile data can actually improve object recognition performance.
3) An interesting phenomenon could be discovered, i.e., the PMVL-based methods (e.g., "UEAF" and "CPMNET") perform worse than Vision and Touch on the GelFabric and LMT datasets.

This phenomenon occurs because "UEAF" and "CPMNET" do not think about big gap (e.g., data format and frequency) between visual and tactile modalities, which inevitably leads to performance degradation.

Moreover, we plot the performance with different missing rate ($\mathrm{MR} = [0.1, 0.2, 0.3, 0.4, 0.5]$) on all the three datasets in terms of accuracy, precision, and F1-Score, respectively, as shown in Figs. 7–9. With increasing missing rate, the performance degradation of our PVTF framework is much smaller than the competitive methods. Taking the GelFabric dataset for example, Ours obtain the accuracy of 97.48% and 71.01% when the missing rate is 0.1 and 0.5, respectively,

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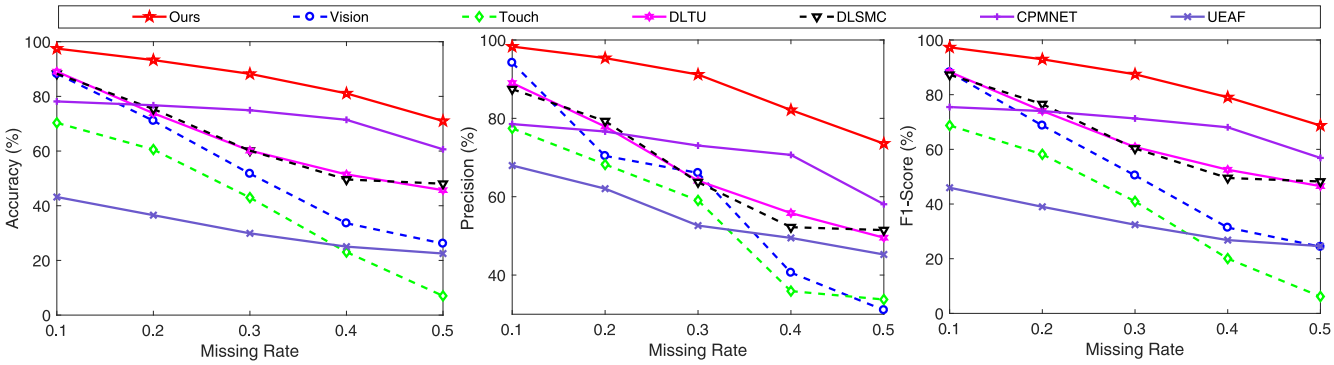

Fig. 7. Object recognition performance (%) on the GelFabric dataset with different missing rate in terms of accuracy, precision, and F1-Score.

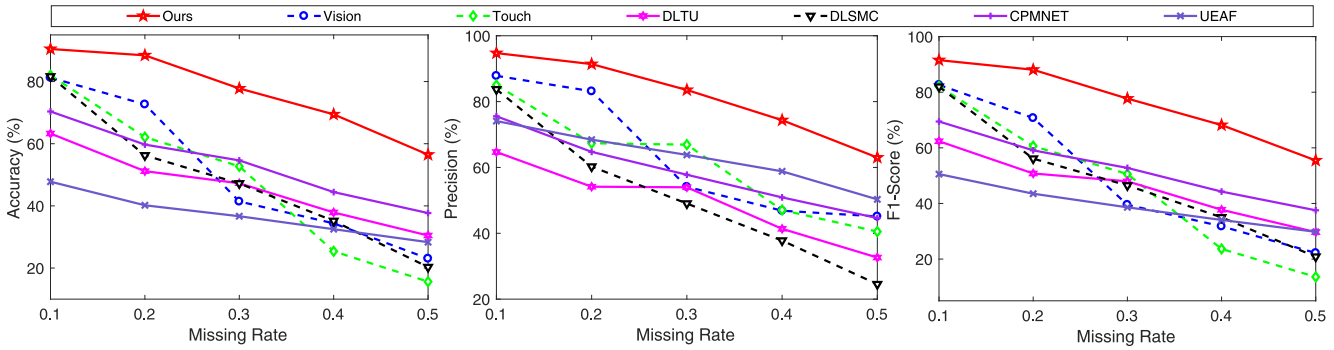

Fig. 8. Object recognition performance (%) on the LMT dataset with different missing rate in terms of accuracy, precision, and F1-Score.

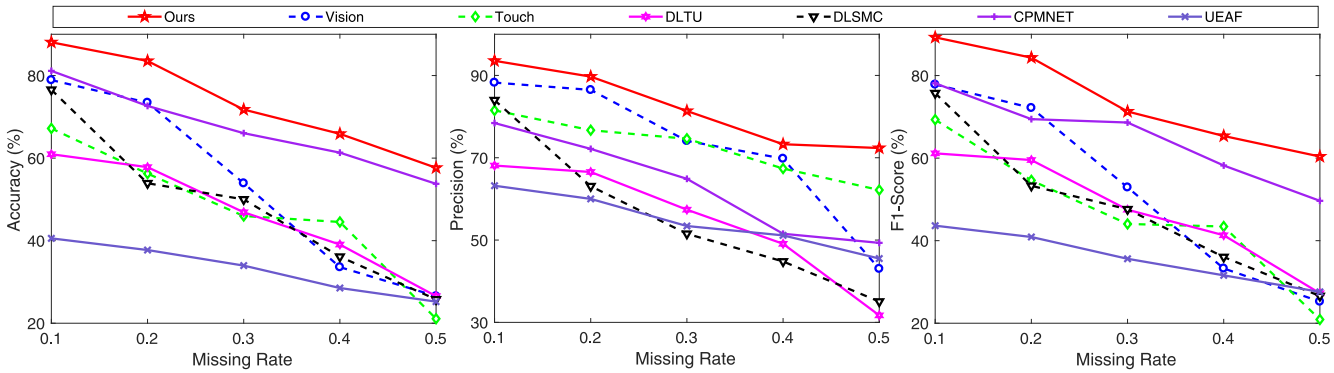

Fig. 9. Object recognition performance (%) on the PHAC-2 dataset with different missing rate in terms of accuracy, precision, and F1-Score.

as shown in Fig. 7. With the increasing missing rate from MR = 0.1 to MR = 0.5, Ours accuracy performance decline is less than 26.47% while the second best competitive methods (i.e., DLTU) performance decline is more than 43.36%. Although, on the GelFabric, LMT, and PHAC-2 datasets, the performance of CPMNET and UEAF degrades slower than Ours, our performance is much better than them in most cases. Thus, we safely draw the conclusion that our PVTF framework has better ability to deal with the missing data situations.

C. Ablation Study

We first investigate the effectiveness of the proposed encoders (i.e., MS encoders and cycle encoder) and MGM network. Then the parameter sensitivity and convergence study are provided in this section.

1) Effectiveness of the Cycle Encoder and MGM Network: To prove the effectiveness of the cycle encoder and MGM network, denoted as "CE" and "MGM" respectively, we utilize several settings to evaluate their effectiveness with object recognition tasks. As shown in Fig. 10, "None" denotes that both the "CE" and "MGM" modules are not used, while Ours denotes all the modules are used; "CE" denotes that only the cycle encoder is used, and "MGM" denotes only the MGM network is used. It is clear that Ours performs the best, which reveals that both the "CE" and "MGM" modules have a positive effect on recognition performance. To be more specific, "CE" achieving better performance than None denotes that the presented cycle encoder module can implicitly learn the complete latent representations of the missing data, which further contributes to improving the object recognition performance. "MGM" achieving better performance

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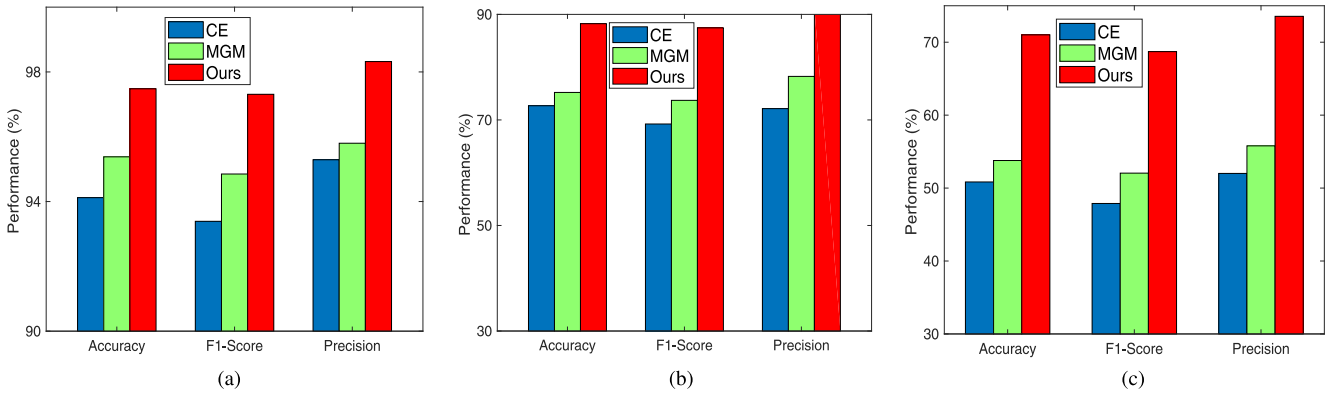

Fig. 10. Accuracy, F1-Score and Precision performance (%) of different missing rate MR on the GelFabric dataset. (a) MR = 0.1. (b) MR = 0.3. (c) MR = 0.5. "None" denotes that none of the modules are used, and Ours denotes that all of the modules are used. "CE" denotes that only the cycle encoder module is used, and "MGM" denotes that only the MGM network is used.

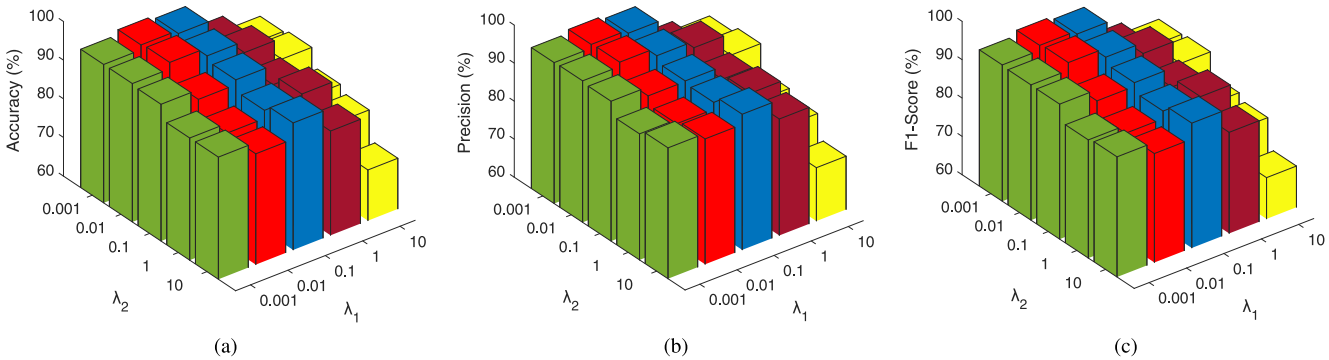

Fig. 11. Object recognition performance (%) with different tradeoff parameters λ_1 and λ_2 in terms of different metrics, i.e., (a) accuracy, (b) precision, and (c) F1-Score on the GelFabric dataset, when the missing rate is set to 0.1.

than None proves its effectiveness of mitigating the visual–tactile gap.

2) Parameter Sensitivity Analysis: In this article, the tradeoff parameters λ_1 and λ_2 should be set appropriately. On the one hand, too large tradeoff parameters could lead to the ability reduction for feature extraction of the MS encoders $E_1(\cdot)$ and $E_2(\cdot)$. On the other hand, too small tradeoff parameters would lead to the visual–tactile gap not been well mitigated. The parameters are tuned from the set {0.001,0.01,0.1,1,10}, as shown in Fig. 11. We notice that Ours achieves the best performance in terms of accuracy, F1-Score, and precision on the GelFabric dataset, when both λ_1 and λ_2 are set to 0.01. Thus, both λ_1 and λ_2 are set to 0.01 as default in this article.

3) Effectiveness of the Modality-Specific Encoders: We compare the proposed MS encoders with several classical network structures, i.e., "AlexNet" [42], "VE1," "VE2." Since the visual and tactile encoders $E_1(\cdot)$ and $E_2(\cdot)$ have similar network structure, we only analyze the visual encoder in this article. As shown in Fig. 12, Ours achieves the best performance for the following reason: the data of the used GelFabric visual–tactile dataset is limited, which leads to too deep or too large networks (i.e., "AlexNet" or "VE2") not be well trained. In addition, Ours achieves better performance than "VE1," which denotes that too small networks also do not have enough capability of high-dimensional data feature extraction. To this end, we design a 5-layer network structure and set the number of network layers and layer sizes as shown in Fig. 4.

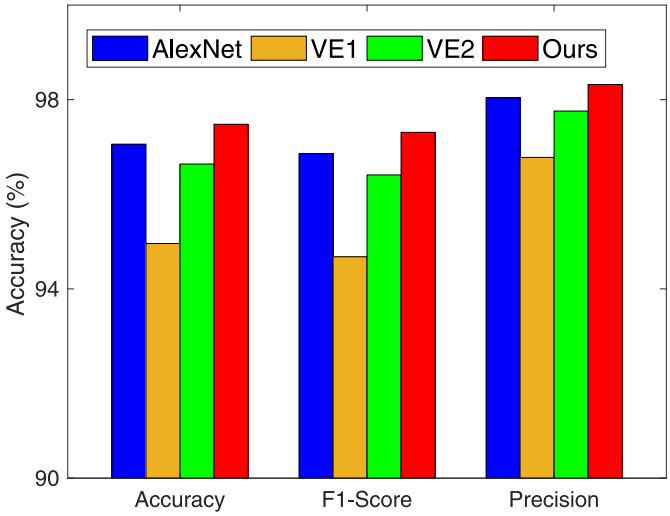

Fig. 12. Accuracy, F1-Score, and precision performance (%) of the visual encoders with different network structures, when the missing rate is set to 0.1. "AlexNet" denotes that we use the AlexNet [42] as the visual encoder. "VE1" denotes that we modify the original visual encoders $E_1(\cdot)$, where the number of layers in the first convolution layer is reduced by half. "VE2"denotes that we modify the original visual encoders $E_1(\cdot)$, where the number of layers in the first convolution layer is doubled.

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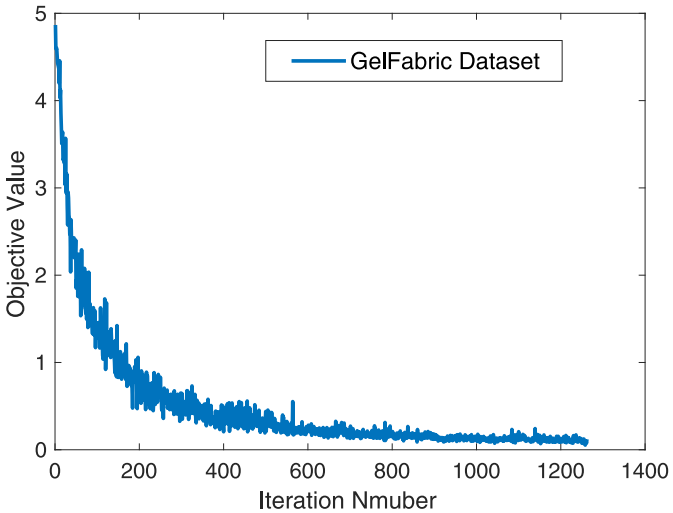

Fig. 13. Convergence analysis of our proposed PVTF framework on the GelFabric dataset, when the missing rate is set to 0.1.

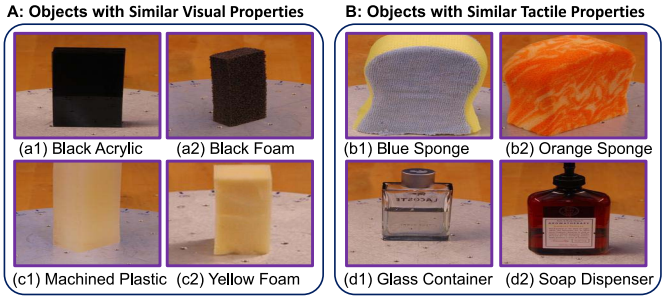

Fig. 14. Representative objects that are used to verify what complementary information can be explored.

4) Convergence Analysis: We introduce the convergence analysis about our proposed model in this section. Due to the length of this article, we only plot the objective value of the overall loss $\mathcal{L}_{\text{total}}$ on the GelFabric dataset in Fig. 13. The object value curves on the remained datasets are similar to this. As shown in Fig. 13, our framework gradually achieves convergence after about 1200 iterations. In this article, we set 1200 as the default maximum number of iterations, i.e., the MaxIter is set to 1200.

5) What Complementary Information Can Be Explored: We do experiments on the PHAC-2 dataset to verify what complementary information can be explored, since the PHAC-2 dataset contains most of the visual and tactile properties of real-world objects. We can see two group representative objects from the PHAC-2 dataset, as shown in Fig. 14, i.e., A: objects with similar visual properties and B: objects with similar tactile properties. It is worth noting that only representative images were selected in Fig. 14. More specifically, in group A, a1, a2 and c1, c2 have similar visual properties (i.e., global shape and color), while their tactile properties (i.e., *hardness, softness, and texture*) are quite different. On the contrary, in group B, b1, b2 and d1, d2 have similar tactile properties while their visual properties are totally different. Thus, we can draw the conclusion that hardness, softness, and texture are the complementary information for visual modality, and global shape and color are the complementary information for tactile modality.

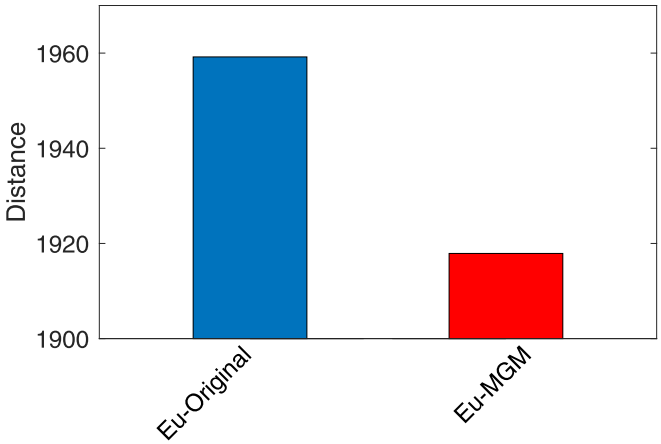

Fig. 15. Euclidean distance comparison of the original visual and tactile feature distribution and the MGM visual–tactile feature distribution. "Eu-Original" and "Eu-MGM" denote the Euclidean distance between original visual and tactile feature distribution, and the Euclidean distance between the visual and tactile feature distribution after gap mitigation, respectively.

6) Visualization of Feature Distributions for Visual and Tactile Modality: To better understand the gap between visual and tactile modalities, we provide visualization figures about visual and tactile feature distributions in Fig. 16. As shown in these figures, the distributions of visual and tactile feature are quite different.

In addition, the distribution difference is reduced after the gap mitigation module. To be more intuitive, we also calculate the Euclidean distance between the visual and tactile feature distributions, as shown in Fig. 15. We can find that the distance of visual and tactile feature distributions is reduced after MGM network, which can further improve the model performance. This phenomenon also proves the effectiveness of our proposed MGM network.

7) Necessity of Cross Modalities: As shown in Fig. 17, we conduct modality ablation experiments to further validate the necessity of cross modalities on the GelFabric dataset. "PVTF" performs the best, which proves the necessity of cross modalities. It indicates that introducing cross modal data could effectively improve the performance of the proposed PVTF framework.

8) Data Loss in Visual Modality and Tactile Modality Is Not Simultaneous: To prove the superiority of our PVTF framework, we conduct experiments when data loss in visual modality and tactile modality is not simultaneous. In this setting, we make the input tactile data complete and let the input visual data to be missing. As shown in Fig. 18, we find that Ours also achieves the best performance, which also demonstrates that our proposed method can outperform either visual or tactile object recognition methods.

9) More Real-World Experiments: To illustrate that our proposed method is more suitable than these previous methods to practice in real-world applications, we do the following two experiments.

We first add idea baseline, when there is no data missing cases, to support the contribution that reasonable processing of the gap between visual and tactile data can actually improve

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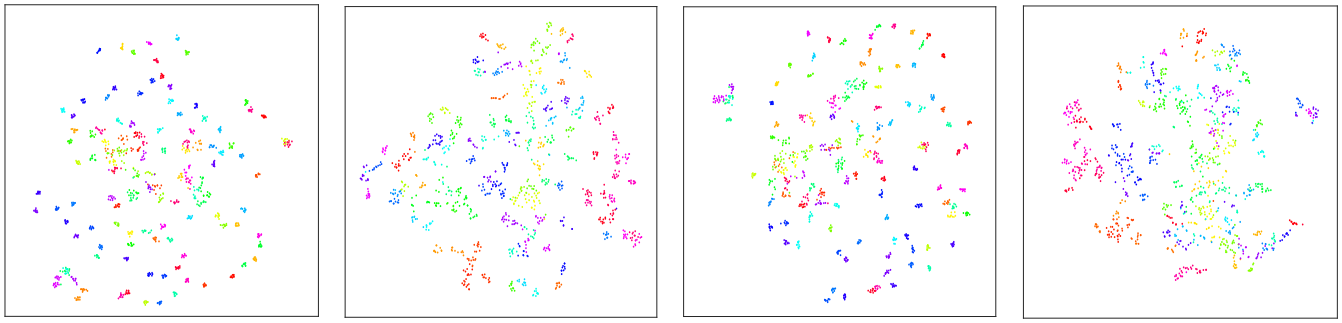

(a) (b) (c) (d)

Fig. 16. Visualization of the visual and tactile feature distributions on the GelFabric dataset, where "MGM" denotes the gap of the feature is mitigated by the proposed MGM network. (a) Visual feature. (b) Tactile feature. (c) MGM visual feature. (d) MGM tactile feature.

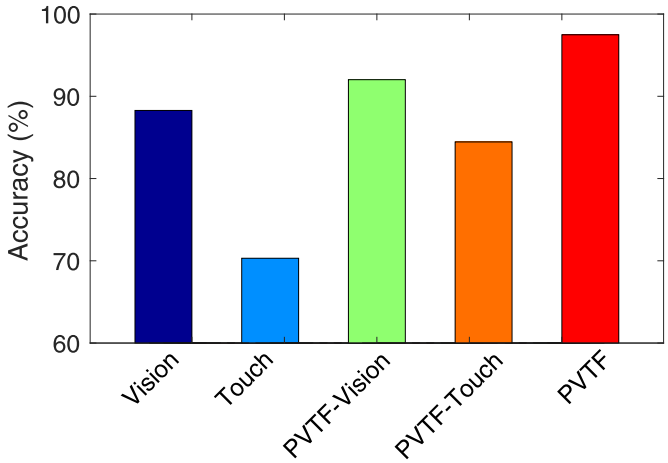

Fig. 17. Accuracy (%) performance of the necessity of cross modalities, which is conducted on the GelFabric dataset. "PVTF" is our proposed model, and "PVTF-Vision" only utilizes the visual data, and "PVTF-Touch" only utilizes the tactile data. Vision and Touch are the single modality-based methods.

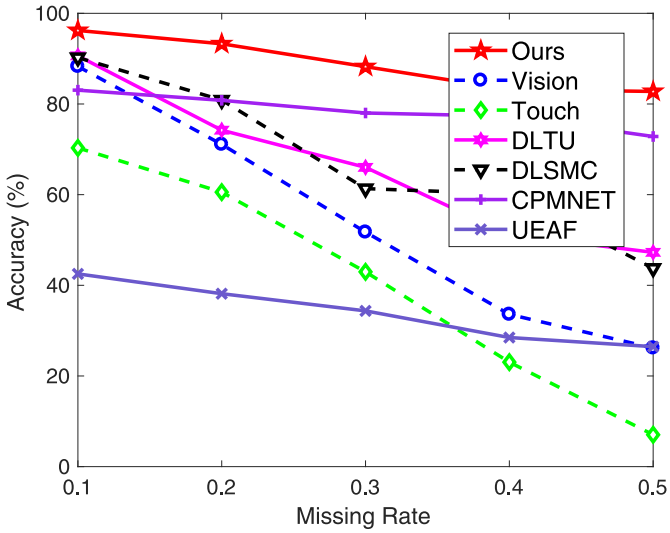

Fig. 18. Accuracy performance (%) of data loss in visual modality and tactile modality is not simultaneously on the GelFabric dataset.

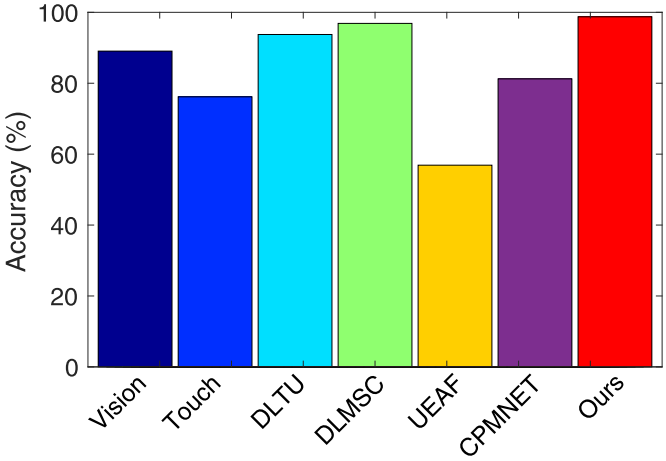

Fig. 19. Accuracy performance (%) on the GelFabric dataset, when there is no missing data both on the visual and tactile modality.

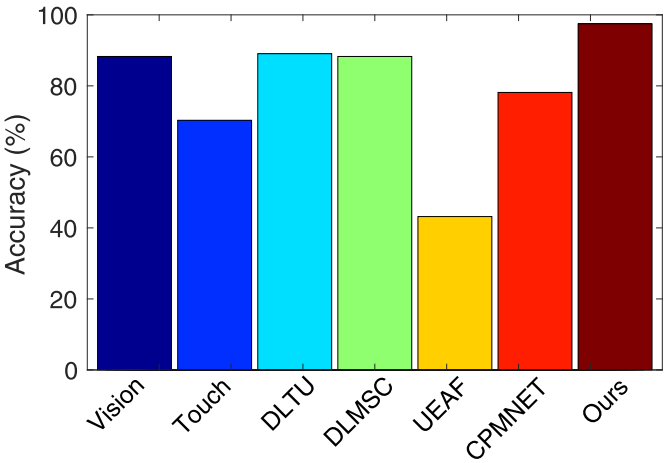

Fig. 20. Accuracy performance (%) on the GelFabric dataset, when the missing rate is set to be 0.1. The missing data points are replaced by average value of the whole dataset, rather than random noises.

the model performance. As shown in Fig. 19, for all the comparison methods and our method, the input data are complete. In this setting, the main difference between our approach and these competitive approaches is that our proposed methods could mitigate the gap between visual and tactile modality better. It reveals that not ignoring the visual–tactile gap can improve object recognition performance.

In addition, in real-world applications, when we encounter data missing situations, an alternative way is to replace the missing data point by the average value of the whole dataset, rather than replaced by random noises. In this setting, as shown in Fig. 20, our proposed PVTF framework also performs

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the best, which further verifies its superiority in real-world applications.

V. Conclusion

In this article, we proposed a new PVTF framework for robotic object recognition, which could tackle the missing input data cases. To be more specific, two MS encoders are employed to project the original partial visual and tactile data into visual subspace and tactile subspace. With an MGM network, the modality-invariant high-level label information is explored to generate gap loss, which helps updating the MS encoders. Then, the visual and tactile subspaces are generated with the regularization of the gap loss, where the gap between visual and tactile modalities is mitigated. Finally, we propose a cycle learning technique to learn a complete subspace, where the missing data are completed and the complementary visual–tactile information is maximized simultaneously. It generates complete latent representations that contain the fused visual–tactile information by a cycle encoder, which attempts to reconstruct incomplete visual and tactile subspaces from the complete subspace. Abundant experiments carried on three visual–tactile fused datasets demonstrate the effectiveness of our PVTF framework, in comparison with state-of-the-art VTFS and PMVL methods.

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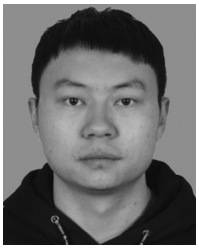


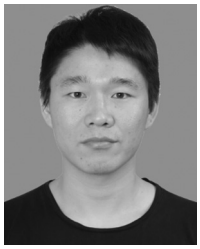
Tao Zhang received the B.S. degree from the Wuhan University of Technology, Wuhan, China, in 2016. He is currently pursuing the Ph.D. degree with the State Key Laboratory of Robotics, Shenyang Institute of Automation, Chinese Academy of Sciences, Shenyang, China, and the University of Chinese Academy of Sciences, Beijing, China.

He has some top-tier conference papers accepted at AAAI, and some top-tier journal papers accepted at TCYB, TSMC-A, and TCSVT. His current research interests include tactile sensing, visual-tactile fusion, robotics, and multiview clustering.

Yang Cong (Senior Member, IEEE) received the B.Sc. degree from Northeast University, Shenyang, China, in 2004, and the Ph.D. degree from the State Key Laboratory of Robotics, Chinese Academy of Sciences, Shenyang, in 2009.

He is a Full Professor with the Chinese Academy of Sciences. He was a Research Fellow of the National University of Singapore, Singapore, and Nanyang Technological University, Singapore, from 2009 to 2011, respectively, and a Visiting Scholar of the University of Rochester, Rochester, NY, USA. He has authored over 80 technical papers and 20+ patents. He was the PI of more than 15 projects, including NSFC Projects, and the National Science and Technology Major Project. His current research interests include robot vision, robot learning, big data, and multimedia and medical image analysis.

Prof. Cong won the National Science Fund for Excellent Young Scholars, the First Prize of Natural Science Award of Liaoning Province, and the First Prize of Natural Science Award of Chinese Association of Automation. He has served on the editorial board of the IEEE TRANSACTION ON CIRCUITS AND SYSTEMS FOR VIDEO TECHNOLOGY, *ACTA Automatica Sinica*, and *Journal of Robotics*.



Jiahua Dong received the B.S. degree from Jilin University, Changchun, China, in 2017. He is currently pursuing the Ph.D. degree with the State Key Laboratory of Robotics, Shenyang Institute of Automation, Chinese Academy of Sciences, Shenyang, China, and the University of Chinese Academy of Sciences, Beijing, China.

He also has some top-tier conference papers accepted at CVPR, ICCV, ECCV, and AAAI. His current research interests include transfer learning, object detection, robotic vision, and medical image processing.

Dongdong Hou received the B.Sc. degree from the Hebei University of Technology, Tianjin, China, in 2014. She is currently pursuing the Ph.D. degree with the State Key Laboratory of Robotics, Shenyang Institute of Automation, Chinese Academy of Sciences, Shenyang, China, and the University of Chinese Academy of Sciences, Beijing, China.

Her current research interests include anomaly detection, deep learning, and sparse representations.

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